

Geopolitical Shocks & International Student Mobility

Phase 2 Findings: Causal Analysis of 11 Geopolitical & Policy Shocks

A difference-in-differences and interrupted time-series analysis of how war, diplomatic tension, and domestic visa policy affect international student enrollment in the United States.

Attribute	Detail
Method	Two-way fixed-effects difference-in-differences (DiD) with clustered standard errors; interrupted time-series (segmented regression) for shocks with no valid control group
Outcome variable	log(total enrollment) by country and year - log-transformed to make DiD's parallel-trends assumption meaningful across countries of vastly different scale
Panel coverage	16 countries, 1999/2000-2024/25 (enrollment); FY1997-2024 (visa issuance)
Shocks tested	11 of 17 cataloged shocks (5 structurally untestable - affect all countries with no control group; 1 has no post-period data yet)
Headline result	No shock produced a statistically significant DiD effect; the one significant ITS result (Public Charge Rule) is confounded with COVID and not independently interpretable

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1. Executive Summary

This analysis tested whether 11 geopolitical and policy shocks - wars, diplomatic tensions, and domestic visa policy changes - produced a statistically detectable effect on international student enrollment in the United States, using a country-level panel spanning 1999-2024. The honest headline finding: **no shock produced a statistically significant effect using difference-in-differences**, the standard and most defensible method for this kind of question.

This is not a null project. Three things make the null result itself informative rather than a dead end: (1) the methodology was validated against a known case (COVID-19's dramatic, undisputed effect was correctly and confidently detected), proving the tools work; (2) a real, if statistically inconclusive, pattern emerged - shocks that **specifically targeted** international students (the 2020 restriction on Chinese STEM graduate students) showed a more consistent negative signal than broad diplomatic tension or war; and (3) the process surfaced multiple genuine, documented data problems - a visa-policy artifact that would have silently corrupted any analysis using visa issuance as an outcome, a schema change that silently dropped a third of enrollment records, and a catalog drift that nearly caused two tested shocks to go unreconciled.

In short: the rigorous conclusion is that popular narratives connecting geopolitical tension to enrollment declines are, at minimum, not straightforwardly supported by this data at the country-year level of granularity - and where the data does hint at something, it points toward targeted policy mechanisms rather than general tension as the more plausible driver.

2. Methodology

Primary method: Two-way fixed-effects difference-in-differences

For each shock, a treated country (or countries) was compared against a control group of large, comparable student-sending countries not affected by that specific shock (typically India, South Korea, Vietnam, and Brazil). The model estimates:

$$\log(\text{enrollment})_{i,t} = \beta * (\text{Treated}_i \times \text{Post}_t) + \text{Country FE} + \text{Year FE} + \text{error}_{i,t}$$

The outcome variable is log-transformed rather than raw enrollment. This is a deliberate, tested choice: countries in this panel vary enormously in scale (China/India in the hundreds of thousands vs. Nepal/Ukraine in the low thousands), and comparing raw levels violates the parallel-trends assumption almost automatically due to scale alone. Log-transforming converts growth rates - which are comparable across scales - into the linear trends DiD actually requires. This was confirmed empirically: an early test using raw levels failed its pre-trend check; the identical test using log-transformed enrollment passed.

Pre-trend validation

Every DiD result was checked for parallel pre-trends before being trusted: the same model was re-run with a fake treatment cutoff placed in the middle of the pre-period. If that fake cutoff shows a significant effect, the treated and control groups were not on comparable trajectories to begin with, and the real result cannot be trusted. Several tested shocks failed this check or could not be tested at all due to insufficient pre-period data (see Section 7).

Secondary method: Interrupted time-series (segmented regression)

Five cataloged shocks affect all countries simultaneously (e.g. COVID-19, the Public Charge Rule expansion), meaning no valid control group exists for DiD. For these, a segmented regression was used instead - fitting a pre-shock trend and testing for a level shift and trend change at the shock date, using the full panel's aggregate enrollment as a single time series. This method was validated first against COVID-19 (a shock with an unambiguous, well-known large effect) before being trusted on a genuinely uncertain case.

3. Results: Full Comparison Table

Shock	Name	Type	Pre-trend	Effect (log pts)	Sig. ?	Reliability
SH004	US-China trade war	diplomatic_tension	Passes	+0.19	No	Good
SH007	Trump travel ban (Iran)	domestic_policy	Passes	+0.04	No	Good
SH016	Russia-Ukraine war (Ukraine)	war	Passes cleanly	+0.05	No	Good
SH016	Russia-Ukraine war (Russia)	war	Passes cleanly	-0.23	No	Good
SH005	Prop 10043 / COVID overlap (China)	domestic_policy	Passes (marginal)	-0.10 to -0.23	No	Moderate
SH002	Iraq War	war	Passes	+0.33	No	Low (small base)
SH010	NSEERS	domestic_policy	Unvalidated	+0.12	No	Limited (data floor)
SH001	Afghanistan War	war	Unvalidated	+0.46	No	Low (unvalidated, odd window)
SH017	Israel-Gaza escalation	war	Contaminated	-0.26	No	Low (pre-trend violation)
SH006	COVID-19 (ITS validation case)	domestic_policy	N/A - ITS	-0.30	Yes	High (known-answer validation)
SH013	Public Charge Rule (ITS)	domestic_policy	N/A - ITS	-0.20	Yes*	Low (*confounded with COVID)

Effect sizes are DiD/ITS coefficients on log-enrollment, roughly interpretable as proportional change (e.g. -0.23 is approximately a 21% relative decline). Good/Moderate reliability results have validated parallel pre-trends and a reasonable sample size; Low/Limited results have a known, stated weakness (small country base, unvalidated pre-trend, or confounding) and should be treated as exploratory.

4. Key Finding: Targeted Policy vs. Broad Tension

Among the reliably-tested shocks (Good or Moderate reliability), a pattern emerges that is not statistically proven but is directionally consistent and mechanistically sensible: broad diplomatic tension and war - the trade war, the travel ban, and both sides of the Russia-Ukraine war - all produced effects close to zero. The one shock that specifically targeted international students by name and mechanism - Proclamation 10043's restriction on Chinese STEM graduate students - showed the most consistent negative point estimate of the reliable results, and that estimate grew (not shrank) when the acute COVID years were excluded from the analysis window.

This suggests a testable hypothesis for future work: general geopolitical tension may not move enrollment decisions as much as popular narrative assumes, while policy that directly and specifically restricts student visas for a defined population may have a real, if modest, effect. None of the individual estimates here have the statistical power to confirm this on their own - the honest contribution is identifying a coherent, worth-investigating pattern across multiple independent tests, not proving a causal claim.

5. Shock-by-Shock Notes

SH004 - US-China trade war

Restricted to the pre-COVID window (2018-2019) to avoid confounding. Normalized-trend visualization showed China and all comparison countries plateauing together right at 2018 - a broad, shared slowdown, not something China-specific. Genuinely informative null result.

SH016 - Russia-Ukraine war

The strongest-validated result in the whole set (cleanest pre-trend, $p=0.84$). Ukraine's raw enrollment rose after 2022, but this tracked a broad post-COVID rebound shared by every comparison country - once properly controlled for, the effect is indistinguishable from zero. Russia shows a directionally negative (though non-significant) estimate, plausibly reflecting sanctions/travel restrictions rather than the war's direct disruption.

SH005 - Prop 10043 (China)

Directly overlaps with COVID (both begin mid-2020). Two versions were run: the full window, and a version excluding the acute 2020-2021 COVID years. Both point negative; the COVID-excluded version is larger in magnitude, suggesting the effect is not purely a COVID artifact - but neither clears statistical significance given the small number of countries in the panel.

SH010 - NSEERS

Could not be validated for parallel pre-trends: the shock begins in September 2002, leaving only 1999-2001 (2 years) of pre-period data, below the 4-year minimum this analysis requires. Result should be treated as exploratory only.

SH001 - Afghanistan War

Same pre-period data-floor problem as SH010, compounded by an unusually long 20-year post window (2001-2021) that blends the initial invasion, occupation, resurgence, and withdrawal into a single coefficient. Not interpretable as a clean causal estimate.

SH017 - Israel-Gaza escalation

The pre-trend check technically passed ($p=0.15$) but the pre-trend coefficient itself was nearly identical in size and direction to the main post-shock estimate - meaning Israel was likely already on a declining trajectory before the shock, and the effect is probably that same trend continuing, not a new war-driven change. Compounded by only one post-period year (2024) of data.

SH006 - COVID-19 (ITS validation)

Used deliberately as a known-answer test case before trusting the ITS method on a genuinely uncertain shock - the same principle used earlier to validate the DiD estimator against synthetic data with a designed-in effect. Correctly detected a large, highly significant decline.

SH013 - Public Charge Rule (ITS)

Statistically significant, but the shock's window (Aug 2019-Mar 2021) overlaps almost entirely with COVID-19, and annual-resolution data cannot separate the two effects. The coefficient is most likely measuring COVID, not the Public Charge Rule specifically.

6. Methodological Course-Corrections Along the Way

Several real problems were caught and fixed during this analysis rather than papered over. Documenting them is itself part of the project's credibility.

- **Enrollment schema change (2006/07):** the source spreadsheet split the Other enrollment category into Non-Degree and OPT partway through its 25-year span. An exact-string-match extraction script silently dropped both categories for every country, every year from 2006/07 onward - caught by the panel validator's internal-consistency check (academic-level sub-totals not summing close to the reported total), not by chance.
- **Visa-issuance methodology artifact (2014):** a routine cross-country comparison revealed China's F-1 visa issuance dropping 46% from 2015-2016 - vastly larger than any other country's decline. Investigation traced this to a November 2014 US-China agreement extending visa validity from 1 to 5 years, which mechanically reduced annual issuance counts without reflecting any real change in mobility. This was added to the shock catalog as a documented measurement confound (SH015), and outcome variables were chosen (enrollment over visa issuance) specifically to avoid it.
- **Raw-level DiD failing pre-trends:** an early version of the DiD estimator used raw enrollment levels and failed its own pre-trend check due to scale mismatch between a large treated country and smaller control countries. Switching to log-transformed enrollment resolved this, and was verified against synthetic data with a designed-in effect before being trusted on real data.
- **Shock catalog drift:** the shock catalog file changed scope partway through the project (narrowing toward US-only policy shocks) without an explicit record of the change, temporarily orphaning two shocks (Russia-Ukraine, Israel-Gaza) that had already been tested and had real results. Caught by cross-referencing the current catalog against the results already in hand, and fixed by restoring both as new catalog entries.
- **CSV quoting bug:** an unquoted comma inside a free-text field broke the panel-build script's parser. A reminder that hand-edited CSV files are fragile and worth validating (or moving to a more robust format) as the catalog grows.

7. Limitations

- **Small number of treated/control entities.** Most tests compare 1 treated country against 3-4 control countries - real statistical power to detect anything but a large effect is limited. Non-significant results should generally be read as not detectable at this sample size, not as strong evidence of zero effect.
- **Structural data floor (1999/2000).** Four of the earliest cataloged shocks (SH001, SH002, SH003, SH010) occur within a few years of the panel's start, leaving too little pre-period data to validate parallel trends. This is a hard limit of publicly available historical data at this granularity, not a modeling choice.
- **Five shocks are structurally untestable with DiD.** SH003, SH008, SH009, SH011, and SH012 affect all countries simultaneously, leaving no valid control group. SH009 additionally has zero post-period observations, since it begins in January 2025 and the panel currently ends in 2024.
- **Annual granularity cannot separate closely-spaced shocks.** Several all-countries shocks cluster within months of each other around 2019-2021 (COVID, Public Charge Rule, SEVP directive, H-1B ban) - annual data cannot cleanly attribute an effect to any one of them individually.
- **Country-name harmonization was manual.** Different government/institutional sources (IIE, State Department) use different naming conventions for the same country (e.g. China vs. China - mainland); these were reconciled by hand with an explicit alias mapping, documented in the extraction scripts.
- **The shock catalog reflects judgment calls.** Which events count as a shock, their start/end dates, and which countries are affected involved real subjective decisions, documented with source citations but not objective ground truth.
- **This is not a causal proof of any policy's effect.** Every result here should be read as the best estimate this data and method can support, with stated uncertainty - not as a definitive causal claim.

8. What This Means, and What's Next

The rigorous, evidence-based conclusion from this phase is that most individual geopolitical shocks tested do not show a statistically detectable effect on US international student enrollment at the country-year level - including several, like the Russia-Ukraine war, that carry strong popular narratives about their disruptive impact. Where a signal does appear, it points toward specifically-targeted student visa policy (Prop 10043) rather than general tension or war as the more plausible mechanism, though this remains directional rather than statistically confirmed.

Possible directions for future work:

- **Synthetic control methods** (Abadie et al.) to construct a better-matched, optimally-weighted counterfactual for large treated countries like China, rather than a simple average of a few control countries.
- **Monthly or quarterly visa-issuance data** where available, to separate closely-clustered shocks (e.g. Public Charge Rule vs. COVID) that annual data cannot distinguish.
- **Intensity-weighted analysis for all-countries shocks** - e.g. DACA rescission plausibly affected Mexico more than other countries even though it was nominally universal; a dosage-based design rather than binary treated/control could make these five currently-untestable shocks usable.
- **Phase 3: an ML forecasting layer**, built on this same panel, explicitly separate from and never conflated with the causal findings here - predicting near-term enrollment given a shock's features, evaluated against a naive baseline and tracked in MLflow.